Quadratic Programming

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Overview

Today, we cover:

- Finish linear programming
- Quadratic programming
- Examples of LP and QP in statistics

Announcements

- Final project due

Course evaluations

Strong duality

- We know that at feasible solutions for both dual and primal, the solution to the dual is always greater or equal
- If there is a difference between the largest primal value and the smallest dual value it is called the "**Duality gap**".
    - When are these values equal?

Strong Duality Theorem: If the primal has an optimal solution, then the dual also has an optimal solution and there is no duality gap, i.e.:

$$\text{objective of dual} \;=\; \text{objective of primal.}$$

- Condition for strong duality: the feasible region must be nonempty and bounded.
- The **duality gap** is zero when strong duality holds.

Complementary slackness theorem

The **complementary slackness theorem** provides a necessary condition for optimality in linear programming. It states that:

If x^* and y^* are feasible solutions of primal and dual problems, then x^* and y^* are both optimal if and only if

1. $y^{*T}(b - Ax^*) = 0$
2. $(y^{*T}A - c)x^* = 0$

This theorem is the foundation of the **interior point method** class of LP solvers.

Complementary slackness theorem

Consider a simple LP problem:

$$\begin{aligned} \textit{max} \quad & z = x_1 + x_2, \\ \textit{s.t.} \quad & x_1 + 2x_2 \leq 100, \\ & 2x_1 + x_2 \leq 100, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Its dual problem is:

Complementary slackness theorem

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Its dual problem is:

$$\begin{aligned} \textit{min} \quad & z = 100y_1 + 100y_2, \\ \textit{s.t.} \quad & y_1 + 2y_2 \geq 1, \\ & 2y_1 + y_2 \geq 1, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Complementary slackness theorem

The complementary slackness theorem states:

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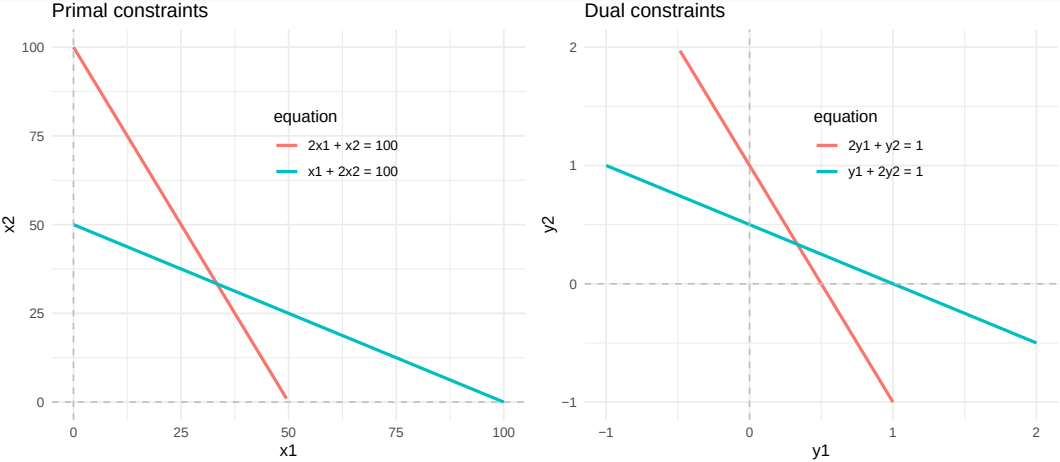
$$
\begin{aligned}
y_1(100 - x_1 - 2x_2) &= 0, \\
y_2(100 - 2x_1 - x_2) &= 0, \\
x_1(y_1 + 2y_2 - 1) &= 0, \\
x_2(2y_1 + y_2 - 1) &= 0.
\end{aligned}
$$

At the optimal point, we have $\mathbf{x} = (\mathbf{100/3}, \mathbf{100/3})$, $\mathbf{y} = (\mathbf{1/3}, \mathbf{1/3})$. The complementary slackness holds.

- All the constraints are bounded in both primal and dual problems.

Plot of constraints



Complementary slackness theorem

Modify the simple LP problem a bit:

$$\begin{aligned} \textit{max} \quad & z = 3x_1 + x_2, \\ \textit{s.t.} \quad & x_1 + 2x_2 \leq 100, \\ & 2x_1 + x_2 \leq 100, \\ & x_1, x_2 \geq 0. \end{aligned}$$

and its dual problem is:

$$\begin{aligned} \textit{min} \quad & z = 100y_1 + 100y_2, \\ \textit{s.t.} \quad & y_1 + 2y_2 \geq 3, \\ & 2y_1 + y_2 \geq 1, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Complementary slackness theorem

Now the complementary slackness theorem states:

$$
\begin{aligned}
y_1(100 - x_1 - 2x_2) &= 0, \\
y_2(100 - 2x_1 - x_2) &= 0, \\
x_1(y_1 + 2y_2 - 3) &= 0, \\
x_2(2y_1 + y_2 - 1) &= 0.
\end{aligned}
$$

For this LP problem, the optimal solutions are $\mathbf{x} = (\mathbf{50}, \mathbf{0})$, $\mathbf{y} = (\mathbf{0}, \mathbf{1.5})$. Complementary slackness still holds. We observe that:

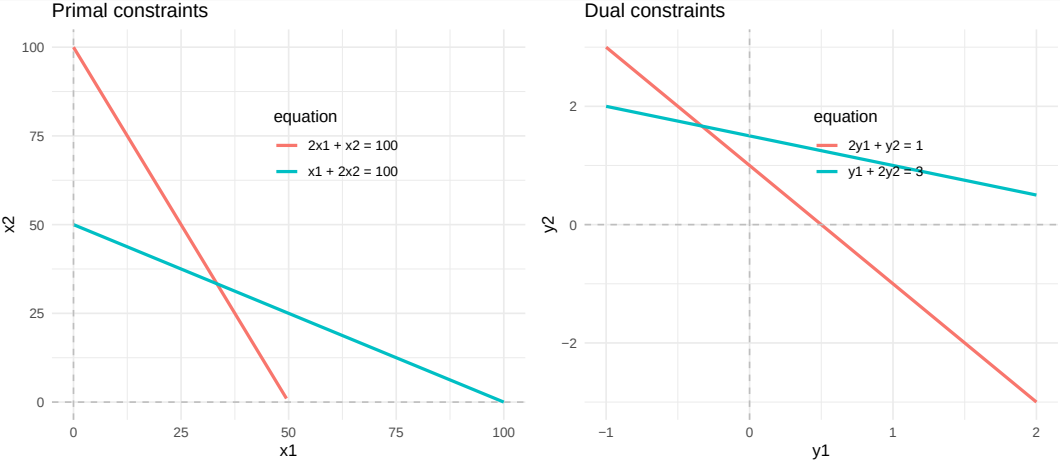
- In the primal problem:
  - The first constraint is **inactive** (slack > 0) at the optimum, so its shadow price $y_1 = 0$
  - Second constraint is **active** (binding, slack $= 0$), so y_2 can be non-zero

Complementary slackness theorem

For this LP problem, the optimal solutions are $\mathbf{x} = (\mathbf{50}, \mathbf{0})$, $\mathbf{y} = (\mathbf{0}, \mathbf{1.5})$. Complementary slackness still holds. We observe that:

- In the primal problem:
  - The first constraint is **inactive** (slack > 0), so its shadow price $y_1 = 0$.
  - The second constraint is **active** (binding, slack $= 0$), so y_2 can be non-zero.
- In the dual problem:
  - The first constraint is **active** (binding), so its primal variable x_1 can be non-zero.
  - The second constraint is **inactive** (slack > 0), so its primal variable $x_2 = 0$.

Plot of constraints



Economics interpretation

Dual variables can be called "**shadow prices**" of the primal constraint, how objective function would increase if constraint was relaxed.

- If a primal constraint is bounded, relaxing that constraint would result in a gain (improve the objective function), shadow price is non-zero.
- If a primal constraint is unbounded, relaxing that constraint would not improve the objective function, shadow price is zero.

Economics interpretation

Bakery example: Imagine you're running a bakery, and can make a limited number of cakes because you have a limited amount of flour. Shadow price tells you how much more profit you'd gain if you had a little more flour.

- If you're tight on flour, getting more flour would let you bake more cakes and make more money. - In this case, the shadow price is nonzero because the extra resource improves your outcome.
- If you have plenty of flour, adding more doesn't help—you're already making as many cakes as you can sell. In this case, the shadow price is zero because extra flour doesn't change your profit.

So, the dual variable (shadow price) tells you how valuable it would be to relax a constraint. Allowing more of that constrained resource. If relaxing the constraint helps, the shadow price is positive. If it doesn't make a difference, the shadow price is zero.

Slack and surplus variables

Both are non-negative variables added to convert an inequality constraint into an equality.

Slack variable: added to a $\leq$ constraint:

$$2x_1 + x_2 \leq 100 \quad \longrightarrow \quad 2x_1 + x_2 + \underbrace{w}_{\geq 0} = 100$$

w absorbs the difference: $w = 100 - 2x_1 - x_2$ (unused capacity). If $w = 0$, the constraint is **binding**.

Surplus variable: subtracted from a $\geq$ constraint:

$$y_1 + 2y_2 \geq 1 \quad \longrightarrow \quad y_1 + 2y_2 - \underbrace{z}_{\geq 0} = 1$$

z measures the excess: $z = y_1 + 2y_2 - 1$ (how far above the bound). If $z = 0$, the constraint is **binding**.

Primal/dual optimality conditions

Goal: collect all conditions for optimality into a single system we can solve numerically.

Step 1: add slack/surplus variables to turn inequalities into equalities.

Primal:

$$\begin{aligned} &\textit{max} \quad cx \\ &\textit{s.t.} \quad \underbrace{Ax + w}_{\text{add slack } w} = b\,, \quad x\,, w \geq 0\,. \end{aligned}$$

Dual:

$$\begin{aligned} &\textit{min} \quad b^T y \\ &\textit{s.t.} \quad \underbrace{A^T y - z}_{\text{subtract surplus } z} = c^T, \quad y\,, z \geq 0\,. \end{aligned}$$

Variable	Size	Meaning
$w = b - Ax \geq 0$	$m \times 1$	primal slack (unused capacity in each constraint)
$z = A^T y - c^T \geq 0$	$n \times 1$	dual surplus (headroom in each dual constraint)

Primal/dual optimality conditions

Step 2: rewrite complementary slackness in matrix form. At optimality, $x_j z_j = 0 \ \forall j$ and $w_i y_i = 0 \ \forall i$.

$$XZe = 0, \quad WYe = 0 \qquad (X, Z, W, Y \text{ diagonal; } e = \mathbf{1})$$

Primal/dual optimality conditions

Why does $XZe = 0$ encode complementary slackness? Suppose $n = 2$:

$$X = \begin{bmatrix} x_1 & 0 \\ 0 & x_2 \end{bmatrix}, \quad Z = \begin{bmatrix} z_1 & 0 \\ 0 & z_2 \end{bmatrix} \implies XZe = \begin{bmatrix} x_1 z_1 \\ x_2 z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This forces $x_j z_j = 0$ for each j. The matrix form stacks all n conditions at once.

Collecting everything: the **optimality conditions** are:

$$\begin{aligned} Ax + w - b &= 0 && \text{(primal feasibility)} \\ A^T y - z - c^T &= 0 && \text{(dual feasibility)} \\ XZe &= 0 && \text{(CS: each } x_j z_j = 0\text{)} \\ WYe &= 0 && \text{(CS: each } w_i y_i = 0\text{)} \\ x, y, w, z &\geq 0 && \text{(non-negativity)} \end{aligned}$$

Primal/dual optimality conditions

Conditions 1–4 give $2n + 2m$ equations in $2n + 2m$ unknowns:

Condition	Equations	Unknowns involved
Primal feasibility	m	x, w
Dual feasibility	n	y, z
CS ($XZe = 0$)	n	x, z
CS ($WYe = 0$)	m	w, y

Can use Newton's method to solve this system.

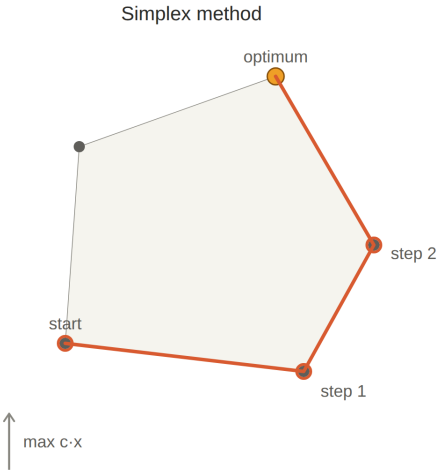
Two strategies for solving LP

	Simplex	**Interior point**
Path	Walks along edges of the feasible region, vertex to vertex	Stays inside feasible region and follows a curved path
Worst for	Exponential (rarely in practice)	Polynomial ~20–50 iterations regardless of problem size
Best for	Small–medium problems	Large-scale problems

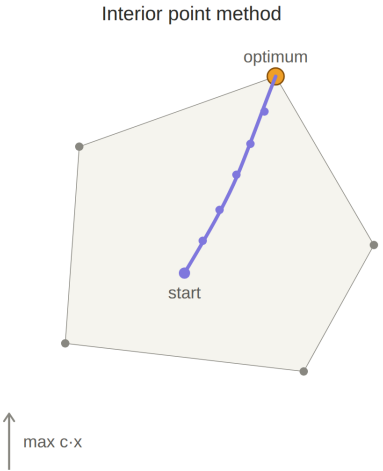
Interior point strategy:

1. Express all optimality conditions as a system $F(x, y, w, z) = 0$
2. Apply Newton's method to drive $F \to 0$
3. At each step, shrink the step size to keep all variables > 0 (stay in the interior)

Two strategies for solving LP



Walks edges, vertex to vertex — few large steps



Cuts through interior — many smooth small steps

Primal-dual interior point method

The primal-dual interior point method finds (x^*, y^*, w^*, z^*) by solving $F = 0$, where F packages all four optimality conditions into one vector function:

$$\mathbf{F}(x, y, w, z) = \begin{bmatrix} \mathbf{A}x + \mathbf{w} - \mathbf{b} \\ \mathbf{A}^T y - \mathbf{z} - \mathbf{c}^T \\ XZe \\ WYe \end{bmatrix} \begin{array}{l} \leftarrow \text{primal feasibility} \\ \leftarrow \text{dual feasibility} \\ \leftarrow \text{CS: each } x_j z_j = 0 \\ \leftarrow \text{CS: each } w_i y_i = 0 \end{array}$$

$F = 0$ is exactly the optimality conditions. We use Newton's method to find the root, modifying each step so all variables stay > 0.

Primal-dual interior point method

Applying Newton's method, if at iteration k the variables are (x^k, y^k, w^k, z^k), we obtain a search direction
$(\delta x, \delta y, \delta w, \delta z)$ by solving the linear equations:

$$F'(x^k, y^k, w^k, z^k) \begin{bmatrix} \delta x \\ \delta y \\ \delta w \\ \delta z \end{bmatrix} = -F(x^k, y^k, w^k, z^k).$$

Here F' is the Jacobian. At iteration k, the equations are:

$$\begin{bmatrix} A & 0 & I & 0 \\ 0 & A^T & 0 & -I \\ Z & 0 & 0 & X \\ 0 & W & Y & 0 \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \\ \delta w \\ \delta z \end{bmatrix} = \begin{bmatrix} -Ax^k - w^k + b \\ -A^T y^k + z^k + c^T \\ -X^k Z^k e \\ -W^k Y^k e \end{bmatrix}$$

Primal-dual interior point method

Then the update will be:

$$(x^{k+1}, y^{k+1}, w^{k+1}, z^{k+1}) = (x^k, y^k, w^k, z^k) + \alpha(\delta x, \delta y, \delta w, \delta z)$$

with $\alpha \in (0, 1]$ chosen so that the result from the next iteration is feasible.

- Solving for $(\delta x, \delta y, \delta w, \delta z)$ using the equation on the previous slide enforces feasibility of solutions

An improved algorithm

The algorithm in its current setup is not ideal because often the steps are small to avoid violating the positivity constraints. Instead:

- The value of $XZe + WYe$ represents the duality gap
- Instead of trying to eliminate the duality gap (set it to 0), reduce the duality gap by some factor in each step

We replace the complementary slackness by:

$$XZe = \mu_x e$$
$$WYe = \mu_y e$$

When $\mu_x, \mu_y \to 0$ as $k \to \infty$ the solution from this system will converge to the optimal solution of the original LP problem. Good selections of μ are $\mu_x^{k+1} = (x^k)^T z^k / n$ and $\mu_y^{k+1} = (w^k)^T y^k / m$

- n and m are the dimensions of x and y

An improved algorithm

Under the new algorithm, at the kth iteration, the Newton equations become:

$$\begin{bmatrix} A & 0 & I & -0 \\ 0 & A^T & 0 & -I \\ Z & 0 & 0 & X \\ 0 & W & Y & 0 \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \\ \delta w \\ \delta z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -X^k Z^k e + \mu_x^k e \\ -W^k Y^k e + \mu_y^k e \end{bmatrix}$$

This provides the **general primal-dual interior point method** as follows:

1. Choose strictly feasible initial solution (x^0, y^0, w^0, z^0), and set $k = 0$. Then repeat (2) and (3) until convergence:
2. Solve system (1) to obtain the updates $(\delta x, \delta y, \delta w, \delta z)$
3. Update the solution $(x^{k+1}, y^{k+1}, w^{k+1}, z^{k+1}) = (x^k, y^k, w^k, z^k) + \alpha^k(\delta x, \delta y, \delta w, \delta z)$. α^k is chosen so that all variables are ≥ 0.

LP in R: simplex vs interior point

Use the following simple LP: $\max\ x_1 + x_2$ s.t.
$x_1 + 2x_2 \leq 100,\ 2x_1 + x_2 \leq 100,\ x_1, x_2 \geq 0$

```r
library(lpSolve)   # simplex method
library(CVXR)      # interior point method (ECOS solver)

obj <- c(1, 1)
A   <- matrix(c(1, 2, 2, 1), nrow = 2, byrow = TRUE)
rhs <- c(100, 100)
```

LP in R: simplex

```r
# lpSolve uses the simplex method
sol_s <- lp("max", obj, const.mat = A,
            const.dir = c("<=", "<="), const.rhs = rhs)

sol_s$solution   # optimal x1, x2
```

```
[1] 33.33333 33.33333
```

```r
sol_s$objval     # optimal objective value
```

```
[1] 66.66667
```

LP in R: interior point

```r
# CVXR calls ECOS: a pure primal-dual interior point solver
x     <- Variable(2, nonneg = TRUE)
prob  <- Problem(Maximize(sum(obj * x)),
                 list(A %*% x <= rhs))
sol_i <- solve(prob, solver = "ECOS")

sol_i$getValue(x)   # optimal x1, x2
```

```
         [,1]
[1,] 33.33333
[2,] 33.33333
```

```r
sol_i$value         # optimal objective value
```

```
[1] 66.66667
```

Introduction to quadratic programming

We have discussed linear programming, where both the objective function and constraints are linear functions of the unknowns.

The **quadratic programming** (QP) problem has a quadratic objective function and linear constraints:

$$\begin{aligned} \max \quad & f(x) = \frac{1}{2}x^T Bx + cx \\ s.t. \quad & Ax \leq b, x \geq 0 \end{aligned}$$

The algorithm for solving QP problem is very similar to that for LP. But first we need to first introduce the **KKT conditions**.

KKT conditions

KKT conditions = complementary slackness, generalized to nonlinear objectives.

- For LP: $\nabla f = c$ (constant) $\Rightarrow$ CS conditions were enough
- For QP/general NLP: $\nabla f(x)$ depends on x $\Rightarrow$ need one extra condition

For a general constrained problem with inequality constraints $g_i(x) \leq 0$ and equality constraints $h_j(x) = 0$, the **KKT conditions** at the optimal x^* are:

Primal feasibility	$g_i(x^*) \leq 0,\ h_j(x^*) = 0$
Dual feasibility	$y_i \geq 0$ (multipliers for $\leq$ constraints are non-negative)
Complementary slackness	$y_i g_i(x^*) = 0$ (inactive constraint $\Rightarrow$ zero multiplier)
Stationarity	$\nabla f(x^*) = \sum_i y_i \nabla g_i(x^*) + \sum_j z_j \nabla h_j(x^*)$

The stationarity condition is new: it says the gradient of f must be expressible as a combination of the constraint gradients — you can't improve f without violating a constraint.

Optimal solution for QP

Apply KKT to the QP problem. Use multiplier y for $Ax \leq b$ and z for $x \geq 0$:

$$L(x, y, z) = \underbrace{\tfrac{1}{2}x^T Bx + cx}_{\text{QP objective}} - \underbrace{y^T(Ax - b)}_{\text{penalty for violating } Ax \leq b} + \underbrace{z^T x}_{\text{penalty for violating } x \geq 0}$$

KKT conditions — same four categories as before, but stationarity now has a Bx term:

Condition	Equation	vs. LP
Primal feasibility	$Ax \leq b,\ x \geq 0$	same
Dual feasibility	$y \geq 0,\ z \geq 0$	same
Complementary slackness	$Y(Ax - b) = 0,\ Zx = 0$	same
Stationarity	$\mathbf{Bx} + c - A^T y + z = 0$	**new: Bx term from quadratic objective**

(Y, Z diagonal.) Solve using interior point — same approach as LP.

Optimal solution for QP

To be specific, define slack variable $w = b - Ax$, the optimality conditions become

$$
\begin{aligned}
Ax - b + w &= 0, \\
Bx + c - A^T y + z &= 0, \\
Zx &= 0, \\
Yw &= 0, \\
x, y, z, w &\geq 0.
\end{aligned}
$$

The unknowns are x, y, z, w. We can then obtain the Jacobians, form the Newton equation and solve for the optimal solution iteratively.

QP in R

The quadprog package provides functions (`solve.QP.compact`) to solve quadratic programming problems.

- Solve $\min \frac{1}{2}(x_1^2 + x_2^2)$ s.t. $2x_1 + x_2 \geq 1$:

```
library(quadprog)
Dmat = diag(rep(1,2))
dvec = rep(0,2)
Amat = matrix(c(2,1))
solve.QP(Dmat=Dmat, dvec=dvec, Amat=Amat, bvec=c(1))
```

Review

We have now covered:

- LP problem set up
- Simplex methods
- Duality
- Interior point algorithms
- Quadratic programming

Now you should be able to formulate an LP/QP problem and solve it. But how are these useful in statistics?

- LP is an optimization algorithm
- There are plenty of optimization problems in statistics, e.g. MLE
- It's just a matter of formulating the objective function and constraints.

Quantile regression

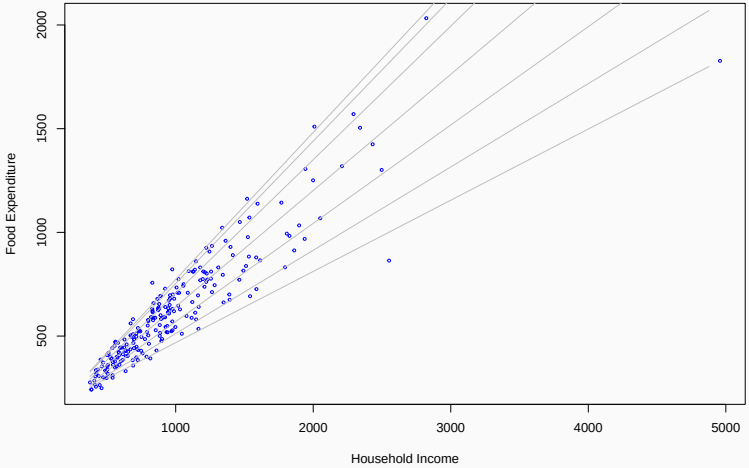
The goal of regression is to understand the relationship between an outcome and covariates. Traditional regression models the conditional mean, $E(Y|X)$

- Relies on normal error assumptions and homoscedasticity
- Sensitive to outliers and skewed distributions

Quantile regression:

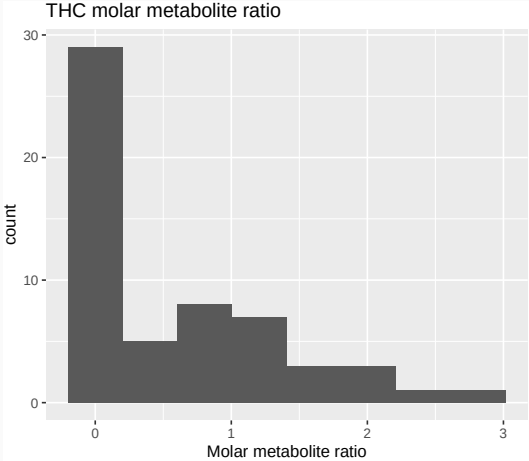
- Provides a more exhaustive description of the data
- The collection of regressions at all quantiles would give a complete picture of outcome-covariate relationships

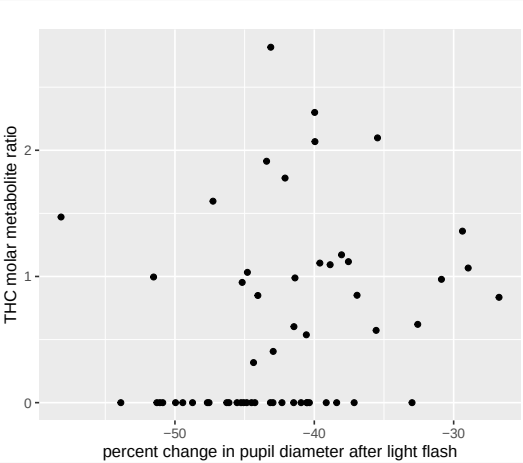
Quantile regression



Colorado cannabis and driving study

Colorado cannabis and driving study





Quantile regression model

Regress conditional quantiles of response on the covariates. Assume the outcome Y is continuous.

- Classical model: $Q_\tau(Y|X) = X\beta_\tau$
- $Q_\tau(Y|X)$ is the τth conditional quantile of Y given X

The above model is equivalent to specifying

$$Y = X\beta_\tau + \epsilon; \quad Q_\tau(\epsilon|X) = 0.$$

In comparison, mean regression is:

$$Y = X\beta + \epsilon; \quad E(\epsilon|X) = 0.$$

Pros and cons

Advantages:

- Regression at a sequence of quantiles provides a more complete view of data
- Inference is robust to outliers
- Allows interpretation in the outcome's original scale of measurement

Disadvantages:

- To be useful, needs to regress on a set of quantiles: computational burden
- Solution has no closed form
- Adaptation to non-continuous outcomes is difficult

The loss function

Link between estimands and loss functions:

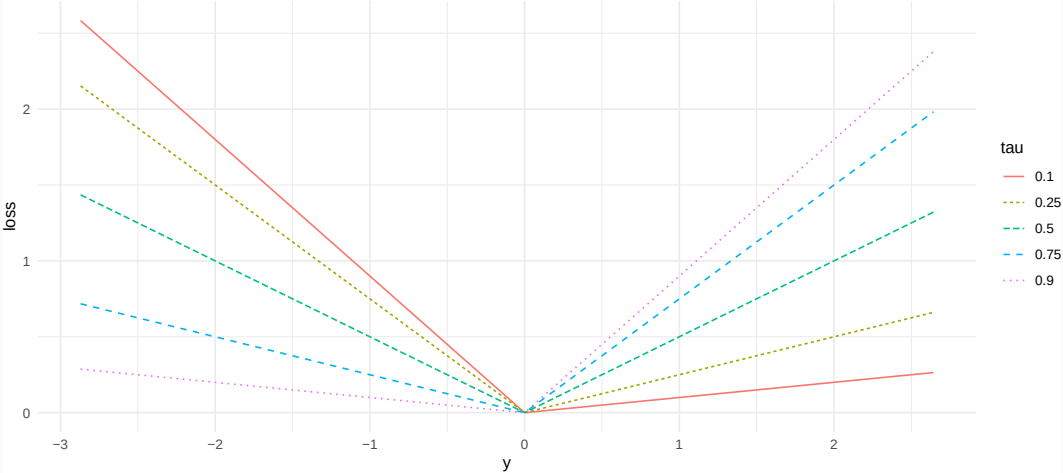
- To obtain sample mean of $\{y_1, y_2, \dots y_n\}$ minimize $\sum_i (y_i - b)^2$
- To obtain sample median of $\{y_1, y_2, \dots y_n\}$ minimize $\sum_i |y_i - b|$

It can be shown that to obtain the sample τth quantile, one needs to minimize asymmetric absolute loss, that is, compute

$$\hat{Q}_\tau = \arg\min_b \left\{ \sum_{i:y_i \geq b} \tau |y_i - b| + \sum_{i:y_i < b} (1-\tau)|y_i - b| \right\}.$$

For notational simplicity, define $\rho_\tau(x) = x[\tau - (x < 0)]$, where $(x < 0)$ is an indicator equal to 1 if $x < 0$ and 0 otherwise. This compactly encodes the asymmetric weighting: $\rho_\tau(x) = \tau x$ when $x \geq 0$ and $\rho_\tau(x) = (\tau - 1)x$ when $x < 0$.

The loss function



Estimator

The linear quantile regression model is fitted by determining

$$\hat{\beta}_\tau = \arg\min_b \sum_{i=1}^n \rho_\tau(y_i - x_i b)$$

The estimator has all "expected" properties:

- Scale equivariance:

$$\hat{\beta}_\tau(ay, X) = a\hat{\beta}_\tau(y, X), \quad \hat{\beta}_\tau(-ay, X) = -a\hat{\beta}_{1-\tau}(y, X)$$

- Shift (or regression) equivariance: $\hat{\beta}_\tau(y + \gamma, X) = \hat{\beta}_\tau(y, X) + \gamma$
- Equivariance to reparametrization of design: $\hat{\beta}_\tau(y, XA) = A^{-1}\hat{\beta}_\tau(y, X)$

Asymmetric Double Exponential (ADE) distribution

Density function for ADE is

$$f(y;\mu,\sigma,\tau) = \frac{\tau(1-\tau)}{\sigma}\exp\left\{-\rho_\tau(\frac{y-\mu}{\sigma})\right\}$$

- Least squares estimator $\iff$ MLE if residuals are Normal
- QR estimator $\iff$ MLE if residuals are ADE
- If residuals re iid $ADE(0,1,\tau)$, then the log-likelihood for β_τ is

$$l(\beta_\tau;Y,X,\tau) = -\sum_i^n \rho_\tau(y_i - x_i\beta_\tau) + c_0$$

QR model fitting

The QR estimation problem $\hat{\beta}_\tau = \arg\min_b \sum_{i=1}^n \rho_\tau(y_i - x_i b)$ can be framed as an LP problem!

- First, define a set of new variables

$$u_i \equiv [y_i - x_i b]_+ \tag{1}$$
$$v_i \equiv [y_i - x_i b]_- \tag{2}$$
$$b_+ \equiv [b]_+ \tag{3}$$
$$b_- \equiv [b]_- \tag{4}$$

Note:

- $[\cdot]_+$ and $[\cdot]_-$ mean the positive and negative part of a number
- $x \equiv [x]_+ - [x]_-$
  - $[x]_+ = \max(x, 0)$
  - $[x]_- = \max(-x, 0)$

QR model fitting

Remember the objective function is $\sum_{i=1}^{n} \rho_\tau(y_i - x_i b)$. Notice that

- When $y_i - x_i b \geq 0$, we have $\rho_\tau(y_i - x_i b) = \tau u_i$, and $v_i = 0$.
- When $y_i - x_i b < 0$, we have $\rho_\tau(y_i - x_i b) = (1-\tau) v_i$, and $u_i = 0$.

So we can write $\rho_\tau(y_i - x_i b) = \tau u_i + (1-\tau) v_i$.

QR model fitting

The optimization problem can then be reformulated as:

$$
\begin{aligned}
\max \quad & -\sum_{i=1}^{n}[\tau u_i + (1-\tau)v_i] \\
\text{s.t.} \quad & y_i = x_i b_+ - x_i b_- + u_i - v_i \\
& u_i, v_i \geq 0, \quad i = 1, \dots, n \\
& b_+, b_- \geq 0
\end{aligned}
$$

This is a standard LP problem can be solved by simplex or interior point methods.

QR model fitting, cont

Written in matrix notation, and make u_i, v_i, b_+, b_- as unknowns, get

$$\begin{aligned} \max \quad & -[\mathbf{0}, \mathbf{0}, \tau, \mathbf{1}-\tau] \begin{bmatrix} \mathbf{b}_+ \\ \mathbf{b}_- \\ \mathbf{u} \\ \mathbf{v} \end{bmatrix} \\ s.t. \quad & [\mathbf{X}, -\mathbf{X}, \mathbf{I}, -\mathbf{I}] \begin{bmatrix} \mathbf{b}_+ \\ \mathbf{b}_- \\ \mathbf{u} \\ \mathbf{v} \end{bmatrix} = \mathbf{y} \\ & \mathbf{b}_+, \mathbf{b}_-, \mathbf{u}, \mathbf{v} \geq 0 \end{aligned}$$

The dual problem is:

$$\begin{aligned} \min \quad & \mathbf{y}^T \mathbf{d} \\ s.t. \quad & \begin{bmatrix} \mathbf{X}^T \\ -\mathbf{X}^T \\ \mathbf{I} \\ -\mathbf{I} \end{bmatrix} \mathbf{d} \geq - \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \tau \\ \mathbf{1}-\tau \end{bmatrix} \\ & \mathbf{d} \text{ is unrestricted} \end{aligned}$$

QR model fitting, cont

Manipulating the dual constraints, get

$$\begin{aligned} \mathbf{X}^T\mathbf{d} &= \mathbf{0} \\ -\tau \le \mathbf{d} &\le 1-\tau \end{aligned}$$

Define a new variable $\mathbf{a} = 1 - \tau - \mathbf{d}$, the original LP problem can be formulated as:

$$\begin{aligned} \max \quad & \mathbf{y}^T\mathbf{a} \\ s.t. \quad & \mathbf{X}^T\mathbf{a} = \mathbf{X}^T(1-\tau) \\ & 0 \le \mathbf{a} \le 1 \end{aligned}$$

Adding slack variables $\mathbf{s}$ for the $\le$ constraints, the problem can be formulated in the standard form, and can be solved by using either Simplex or Interior Point methods.

$$\begin{aligned} \max \quad & \mathbf{y}^T\mathbf{a} \\ s.t. \quad & \mathbf{X}^T\mathbf{a} = \mathbf{X}^T(1-\tau) \\ & \mathbf{a} + \mathbf{s} = 1 \\ & \mathbf{a}, \mathbf{s} \ge 0 \end{aligned}$$

QR model fitting, cont

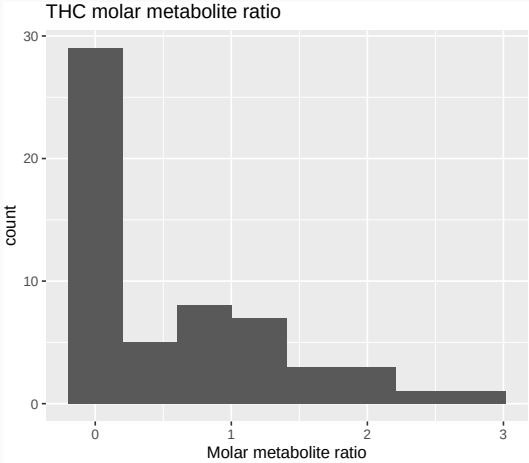
In this form,

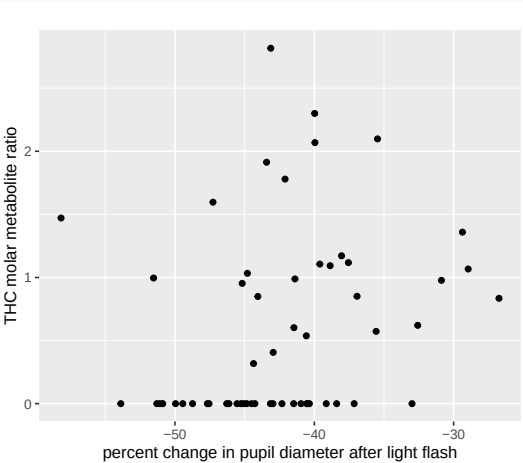
- $\mathbf{y}$ (n-vector) are outcomes
- $\mathbf{X}$ ($n \times p$ matrix) are predictors,
- $\mathbf{a}$ and $\mathbf{s}$ (n-vectors) are unknowns.
- $1 - \tau$ and 1 need to be n-vectors

There are $2n$ unknowns and $p + n$ constraints.

Once we have optimal $\mathbf{a}$, $\mathbf{d}$ can be obtained given τ (the quantile). Then depending on which constraints are hit in the dual problem, one can determine the set of basic variables in primal problem, and then solve for the primal variables (just solve a set of linear equations), and then obtain β.

QR in R


THC molar metabolite ratio
count
30
20
10
0
0
1
2
3
Molar metabolite ratio


THC molar metabolite ratio
2
1
0
−50
−40
−30
percent change in pupil diameter after light flash

QR in R

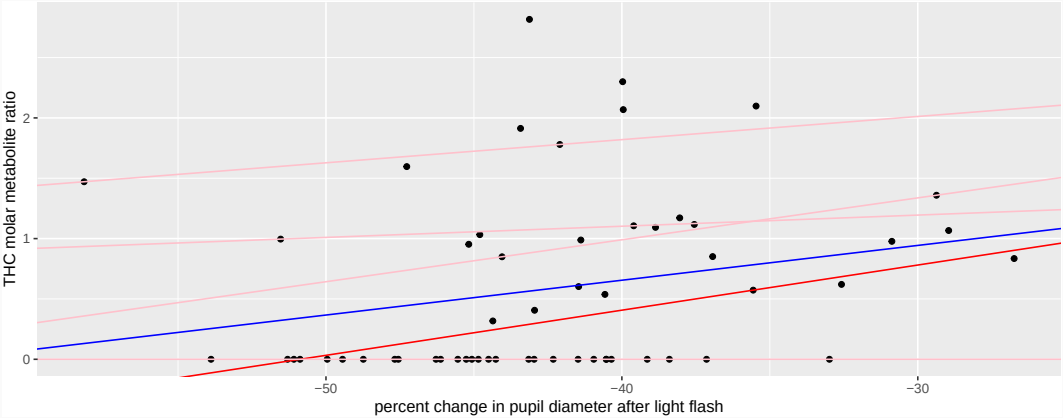
Want to compare linear and quantile regression

```
ols = lm(t_mmr1 ~ p_change, data = cannabis)

# default method, which is a simplex method
median_regression = rq(t_mmr1 ~ p_change, tau = 0.5, data = cannabis,
                       method = "br")
qr.9 = rq(t_mmr1 ~ p_change, tau = 0.9, data = cannabis,
          method = "br")
qr.1 = rq(t_mmr1 ~ p_change, tau = 0.1, data = cannabis,
          method = "br")
qr.7 = rq(t_mmr1 ~ p_change, tau = 0.7, data = cannabis,
          method = "br")
qr.8 = rq(t_mmr1 ~ p_change, tau = 0.8, data = cannabis,
          method = "br")

summary(ols)$coefficients
```

```
              Estimate Std. Error  t value    Pr(>|t|)
(Intercept) 1.80873656 0.65925047 2.743626 0.008188134
```

QR in R

Blue is ols, red is median regression, pink is tau = 0.9



LASSO

Consider usual regression settings with data $(\mathbf{x_i}, \mathbf{y_i})$, where $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})$ is a p-vector of predictors and y_i is the response for the ith subject. The OLS setting finds a coefficient vector to minimize the residual sum of squares:

$$\hat{\beta} = \arg\min_{\beta} \sum_{i}^{n} (y_i - x_i\beta)^2$$

- Solution is the MLE assuming a Normal model:

$$y_i = \mathbf{x}_i\beta + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

- This is undesirable when p is large because

1. When p>n, model can't be fit
2. One typically wants a more parsimonius model for interpretability

LASSO

LASSO stands for "Least Absolute Shrinkage and Selection Operator", which aims for model selection when p is large (works even $p > n$). The LASSO procedure will "shrink" the coefficients toward 0 and eventually force some to be exactly zero

- predictors with $\beta = 0$ will be selected out

The LASSO estimates are defined as:

$$\tilde{\beta} = \arg\min_{\beta} \left\{ \sum_{i}^{n} (y_i - \mathbf{x}_i \beta)^2 \right\} s.t. ||\beta||_1 \leq \lambda$$

- $||\beta||_1 = \sum_{j=1}^{p} |b_j|$ is the L_1 norm
- $\lambda \geq 0$ is a tuning parameter controlling the strength of shrinkage

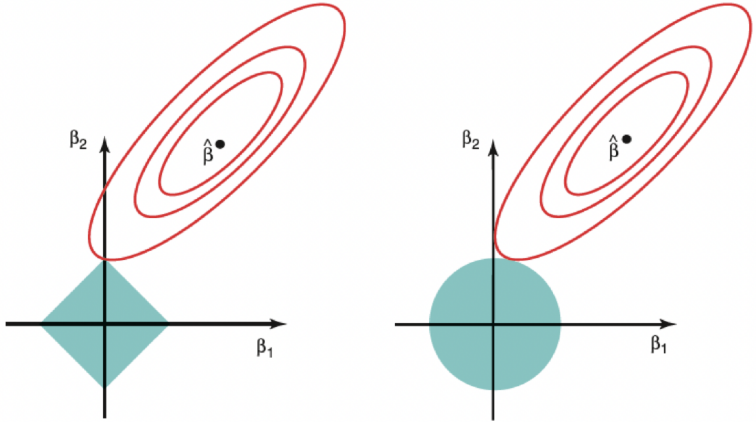
What kind of optimization problem does this look like?

Regularization side note

LASSO is a regularization approach- regularization prevents overfitting by adding a penalty term to the objective function.

- LASSO tries to minimize the residual sum of squares, with a constraint on the sum of the absolute values of the coefficients.
- There are other types of regularized regressions
- **Ridge regression**: adds an L_2 penalty, $\sum_j b_j^2 \leq \lambda$

LASSO vs. Ridge visualized



LASSO model fitting

The LASSO problem can be solved by quadratic programming algorithms.

$$\max \quad -\sum_{i}^{n}\left(y_i - \sum_j \beta_j x_{ij}\right)^2 \tag{5}$$

$$s.t. \quad \sum_j |b_j| \leq \lambda \tag{6}$$

- **Issue**: not in standard LP/QP form, since the constraints have the absolute value operator

LASSO model fitting

The trick is to convert the problem into the standard QP problem setting, i.e., to remove the absolute value operator. This is the **same decomposition used in QR** model fitting (u_i, v_i for residuals), now applied to the coefficients β_j.

- Let $\beta_j = \beta_j^+ - \beta_j^-$, where $\beta_j^+, \beta_j^- \geq 0$ (positive and negative parts)
- Then $|\beta_j| = \beta_j^+ + \beta_j^-$ and the problem can be written as:

$$
\begin{aligned}
\max \quad & -\sum_i^n \left(y_i - \sum_j \beta_j^+ x_{ij} + \sum_j \beta_j^- x_{ij} \right)^2 \\
s.t. \quad & \sum_j (\beta_j^+ + \beta_j^-) \leq \lambda, \\
& \beta_j^+, \beta_j^- \geq 0
\end{aligned}
$$

This is now a standard QP problem can be solved by standard QP solvers.

A little more on LASSO

The Lagrangian for the LASSO optimization problem is:

$$L(,\lambda) = -\sum_{i=1}^{n}\left(y_i - \sum_j \beta_j x_{ij}\right)^2 - \lambda\sum_{j=1}^{p}|\beta_j|$$

This is equivalent to the likelihood function for a Bayesian model with a double exponential (DE) prior on βs (remember ADE used in quantile regression?):

$$Y|X,\beta \sim N(X\beta,\sigma^2)$$
$$\beta_j \sim DE(1/\lambda)$$

where the DE density function is

$$f(x,\tau) = \frac{1}{2\tau}\exp\left(\frac{-|x|}{\tau}\right)$$

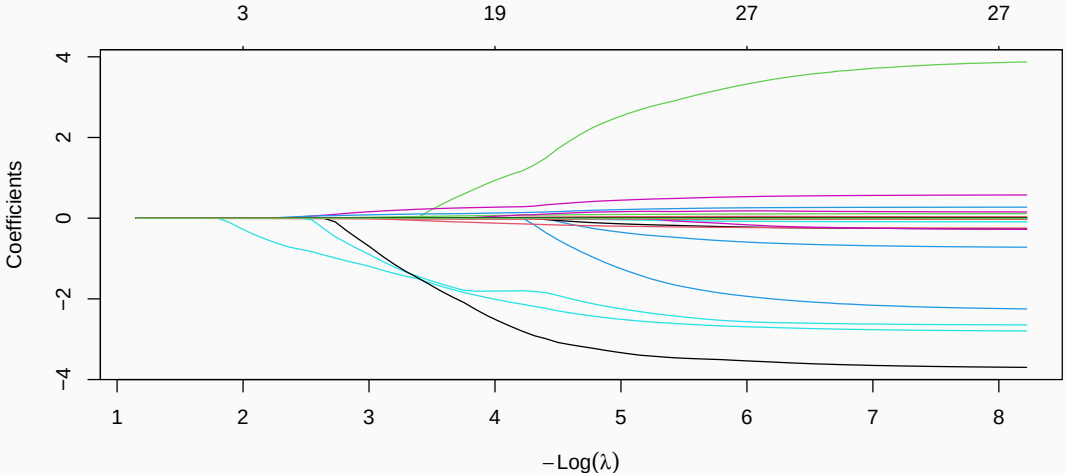
LASSO in R

The package `glmnet` is standard for elastic net and LASSO in R.

```
### Lasso results from R package "glmnet"
library(glmnet)
lasso_mod = glmnet(x = dplyr::select(cannabis, -id, -t_mmr1),
                   y = cannabis$t_mmr1)
plot(lasso_mod, "lambda")
```

LASSO in R

The package `glmnet` is standard for elastic net and LASSO in R.



Resources

- Hao Wu's lecture with matrix representation of Simplex algorithm

Some videos

- convexity explained, and duality
- KKT conditions explained
- Simple explanation of Lagrangian

Lagrange multipliers

The method of Lagrange multiplier is a general algorithm for optimization problems with equality constraints. For example, consider a problem:

$$\begin{aligned} \max \quad & f(x, y) \\ s.t. \quad & g(x, y) = c \end{aligned}$$

We introduce a new variable λ called a **Lagrange multiplier** and form the following new objective function:

$$L(x, y, \lambda) = f(x, y) + \lambda[g(x, y) - c]$$

We will then optimize L with respect to x, y and λ using typical methods. Note that the condition $\partial L / \partial \lambda = 0$ at the optimal solution guarantees that the constraints will be satisfied.

The barrier problem

The interior point algorithm is closely related to the **Barrier Problem**. Go back to the primal problem:

$$\begin{aligned} \max \quad & z = cx \\ s.t. \quad & Ax + w = b; x, w \geq 0 \end{aligned}$$

The non-negativity constraints can be replaced by adding two **barrier terms** in the objective function. The barrier term is defined as $B(x) = \sum_j \log x_j$, which is finite as long as x_j is positive. Then the primal problem becomes:

$$\begin{aligned} \max \quad & z = cx + \mu_x B(x) + \mu_y B(w) \\ s.t. \quad & Ax + w = b \end{aligned}$$

The barrier terms make sure x and w won't become negative.

The barrier problem

The Lagrangian for this problem is (using y as the multiplier):

$$L(x, y, w) = cx + \mu_x B(x) + \mu_y B(w) + y^T(b - w - Ax)$$

The optimal solution for the problem satisfies:

$$c + \mu_x X^{-1} e - A^T y = 0 \tag{7}$$
$$b - w - Ax = 0 \tag{8}$$
$$\mu_y W^{-1} e - y = 0 \tag{9}$$

Define new variables $z = \mu_x X^{-1} e$ and rewrite these conditions, we obtain exactly the same set of equations as the relaxed optimality conditions for primal-dual problem. **Note**: $z = \mu_x X^{-1} e$ is also a condition so there are 4 in total.